rag -- 2026-10-22

5 new paper(s) · topic: Retrieval-augmented generation: retrieval quality, grounding and faithfulness, evaluation benchmarks, and RAG system design.

Worth reading in full

[P1] Retrieval Augmented Generation Compositionality -- P1 provides the most comprehensive systematic analysis of how retrieval, faithfulness, and corpus scale interact across a massive factorial space, offering a reusable framework (LASAGNA) that is directly relevant to all subsequent papers.

Themes

These papers converge on the challenge of evaluating and designing RAG pipelines beyond isolated component gains. LASAGNA in P1 shows that three strong retrieval tricks (cross-encoder reranking, disabling MMR, document-awareness prefixes) predict a +46.6 % lift but only deliver +17.2 % when combined, highlighting severe interaction effects across 450 configurations [P1]. Similarly, P2 demonstrates that scaling corpus size and changing document representation (abstract, flat full-text, TreeRAG hierarchical) can improve MCQA accuracy by up to 9.6 pp, yet the benefits vary with the generator and evidence format, exposing trade-offs between efficiency and performance that are not captured by simple retrieval metrics [P2]. Both studies underscore that retrieval quality (e.g., NDCG@10, Recall@100) does not reliably predict downstream faithfulness or answer accuracy, a point quantified in P1 (Pearson r ~ 0) and echoed in P4 where qualifier-sensitive answer selection (JCA) improves accuracy by 4-11 pp even when candidate recall is held constant, suggesting that downstream reasoning constraints matter as much as raw retrieval scores [P4]. Safety-oriented work (P3) adds another dimension: modest query-editing (C²U) can reduce target Hit@5 by ~4 pp without harming control queries, but the suppression remains partial and the impact on generated answer faithfulness is unmeasured, indicating a gap in evaluating RAG safety alongside relevance [P3]. Finally, P5 introduces a macro-adaptive parallel reasoning framework (MAPS) that curbs query-intent drift and achieves consistent performance gains on multiple QA benchmarks, improving accuracy by 6.8% on Qwen3-32B and 5.8% on Qwen3.6-35B-A3B [P5]. Across the set, a recurring theme is that component-level improvements (retrieval tricks, hierarchical indexing, query edits, parallel search directions) often interact in non-additive ways, and current benchmarks focus on isolated metrics rather than holistic pipeline robustness.

5 theme sentence(s) checked against the paper summaries; 1 corrected, 0 removed.

At a glance

[P1] Retrieval Augmented Generation Compositionality · relevance 0.64
The paper shows that isolated gains of Retrieval-Augmented Generation (RAG) components often disappear when combined, and that optimal pipelines differ across retrieval quality, answer faithfulness, and corpus scale.

[P2] Large-Scale Biomedical RAG: Corpus Scaling, Document Representation, and Efficiency Trade-offs · relevance 0.53
This study systematically evaluates how corpus size, document representation, and reranking affect biomedical Retrieval-Augmented Generation (RAG), finding that larger corpora and hierarchical representations can improve multiple-choice QA accuracy but trade off efficiency, with gains varying by model and evidence format.

[P3] C^{2}U in RAG: Compositional Concept Unlearning in Retrieval-Augmented Generation · relevance 0.49
C^{2}U modestly lowers target retrieval hits (~4 pp) without harming control queries, using a bounded query edit that rotates queries <=0.2 rad.

[P4] Joint Constraint Aggregation for Qualifier-Sensitive Retrieval-Augmented QA · relevance 0.42
Joint Constraint Aggregation improves qualifier-sensitive answer selection in retrieval-augmented QA, especially when retrieval favors a competing near-miss answer.

[P5] MAPS: Macro-Adaptive Parallel Reasoning for Regulating Search Momentum in Retrieval-Augmented Generation · relevance 0.39
MAPS introduces a macro-adaptive parallel reasoning framework that curbs query-intent drift in retrieval-augmented generation and yields consistent performance gains on multiple QA benchmarks.

[P1] Retrieval Augmented Generation Compositionality

Ido Levy, Segev Shlomov, Leshem Choshen · arXiv 2610.29722v1 · https://arxiv.org/abs/2610.29722v1 · relevance 0.64

TL;DR: The paper shows that isolated gains of Retrieval-Augmented Generation (RAG) components often disappear when combined, and that optimal pipelines differ across retrieval quality, answer faithfulness, and corpus scale.

Problem
Current RAG research evaluates techniques (embedding, chunking, retrieval, reranking, etc.) one at a time, leaving it unclear whether their isolated improvements survive in full pipelines, whether the same design choices benefit both retrieval quality and answer faithfulness, and whether findings on small corpora generalize to larger ones [1][3].

Method
The authors introduce LASAGNA, an open-source, stage-isolated framework for controlled RAG evaluation, and run a factorial study of 450 configurations across eight benchmarks with corpora ranging from 5 K to 200 K documents [1][3].

Key results
- Three individually strong retrieval techniques were predicted to give a +47 % combined improvement but achieved only +17 % when used together [1].
- The three largest separately measured retrieval gains (cross-encoder reranking, disabling MMR, document-awareness prefixes) predict a +46.6 % lift over the DEFAULT_STANDARD baseline (NDCG@10 = 0.412) [2]; the BALANCED pipeline that combines them yields only +17.2 % (a 63 % shortfall) [2].
- Adding all five positive retrieval techniques raises the shortfall to 72 %; across 17 tested combinations the shortfall ranges from 7 % to 72 % with a median of 41 % [2].
- Retrieval quality and answer faithfulness are essentially uncorrelated (pooled Pearson r = -0.01, p = 0.91); 5 % of configurations have high faithfulness but low retrieval quality and 6 % the opposite [2].
- Retrieval quality is most associated with embedding, enrichment, and reranking (?² = 0.164, 0.144, 0.090), while faithfulness is driven by query augmentation and agentic gating [2].
- The best component composition changes as the corpus grows; the top-performing pipeline on a small corpus does not remain optimal at larger scales [1][3].

Limitations
The evaluation uses a single constructed English multi-domain test bed (<=93 retrieval queries, <=200 generation queries per experiment) and does not cover multilingual corpora, real enterprise or private data, or open-weight generators [5][8]. Faithfulness is measured with only two proprietary generator models as a sensitivity check, and absolute faithfulness scores rely on a single judge's calibration [5][8]. The benchmarks are limited to five domains, and some experiments used fewer queries than pre-registered due to rate-limit constraints [5][8].

Sources

[1] arXiv 2610.29722v1.pdf, arXiv abstract

[2] arXiv 2610.29722v1.pdf, page 6

[3] arXiv 2610.29722v1.pdf, page 9

[5] arXiv 2610.29722v1.pdf, page 15

[8] arXiv 2610.29722v1.pdf, page 102

[P2] Large-Scale Biomedical RAG: Corpus Scaling, Document Representation, and Efficiency Trade-offs

Adrián Sánchez-Freire, Gonzalo Pérez-Gómez, Aitziber Atutxa, Ander Barrena · arXiv 2610.26339v1 · https://arxiv.org/abs/2610.26339v1 · relevance 0.53

TL;DR: This study systematically evaluates how corpus size, document representation, and reranking affect biomedical Retrieval-Augmented Generation (RAG), finding that larger corpora and hierarchical representations can improve multiple-choice QA accuracy but trade off efficiency, with gains varying by model and evidence format.

Problem
The paper addresses how scaling the retrieval corpus, choosing different document representations (abstract, flat full-text, hierarchical), and applying reranking influence the effectiveness and efficiency of biomedical RAG systems [1].

Method
The authors construct controlled scaling protocols by progressively expanding PubMed and MIRIAD retrieval pools while keeping gold documents constant, creating five corpus sizes plus an extra "full" PubMed setting of ~4.3 M documents [2]. They compare three representations: single-abstract units, overlapping full-text chunks (<=1,600 chars), and TreeRAG hierarchical units built from the same full-text sources [2]. Retrieval quality is measured with Macro Recall@100 (first-stage) and Recall@5 after reranking the top 100 candidates; end-to-end performance is assessed via multiple-choice question-answering (MCQA) accuracy for LLM-only, RAG (top 5 evidence units), and Oracle (gold evidence) conditions [3]. Efficiency is quantified by query latency and index storage, reporting that hierarchical indexing uses 67 % fewer indexed units and 44-56 % less storage than flat full-text [1].

Key results
- RAG improves three of four model configurations on five external biomedical benchmarks, with mean accuracy gains up to **9.6 percentage points**; on a held-out test set, all four configurations improve by **10.2-30.3 points** [1].
- The strongest per-generator gain is achieved by **Gemma-3-4B**, which sees an accuracy increase of **27.50 percentage points** in its best configuration [4].
- Hierarchical (TreeRAG) representation matches or outperforms flat full-text retrieval in most end-to-end tests while requiring **67 % fewer indexed units** and **44-56 % less storage** [1].
- Abstract-level retrieval yields the best results when the reference evidence is abstract-aligned (e.g., MedAESQA), whereas full-text and hierarchical representations give larger gains on external benchmarks that use richer evidence [5].
- Reranking's benefit depends on corpus and retrieval setting; it is applied only to MIRIAD experiments (using Qwen3-Reranker-8B) and omitted for PubMed experiments (which retain the original Qwen3-Embedding-8B ranking) [4].

Limitations
The evaluation is limited to multiple-choice QA, using a small MedAESQA set and a synthetically constructed MIRIAD dataset that shares construction with its test split, so results may not generalize to open-ended generation, explanation quality, evidence citation, or clinical validity [6].

Sources

[1] arXiv 2610.26339v1.pdf, arXiv abstract

[2] arXiv 2610.26339v1.pdf, page 3

[3] arXiv 2610.26339v1.pdf, page 4

[4] arXiv 2610.26339v1.pdf, page 5

[5] arXiv 2610.26339v1.pdf, page 10

[6] arXiv 2610.26339v1.pdf, page 11

[P3] C^{2}U in RAG: Compositional Concept Unlearning in Retrieval-Augmented Generation

Yiyang Duan, Fan Wu, Tiantong Wu, Che Wang et al. · arXiv 2610.29223v1 · https://arxiv.org/abs/2610.29223v1 · relevance 0.49

TL;DR: C^{2}U modestly lowers target retrieval hits (~4 pp) without harming control queries, using a bounded query edit that rotates queries <=0.2 rad.

Problem
Retrieval-augmented generation can surface unwanted concepts through the retrieved evidence even when the downstream generator is moderated, creating safety risks. [1]

Method
Compositional Concept Unlearning (C^{2}U) leaves the encoder and document index unchanged and instead edits the raw query by adding a bounded "edited-minus-original" delta derived from a profile-driven direction routed through an augmented atom bank; the edit rotates the query by at most 0.2 rad before retrieval. [1][3]

Key results
- Across five datasets, four encoders and three random seeds, the selected C^{2}U configuration reduces pooled binary target Hit@5 from 94.73 % to 90.48 % (a 4.25 percentage-point drop) while incurring non-positive control costs on every dataset-condition cell. [2][3]
- Per-dataset single-concept suppression (Hit@5 drop, control-neighbor change CN, target-exposure change TE) includes: Amazon +6.33 pp H5 Drop, -0.64 CN, -12.44 TE; ArXiv +5.67 pp H5 Drop, -1.25 CN, -7.64 TE; IMDBd +2.58 pp H5 Drop, 0 CN, 0 TE; MNLI +3.42 pp H5 Drop, -1.08 CN, -6.03 TE; TOFU +3.25 pp H5 Drop, -0.44 CN, -2.14 TE. [2]
- The pooled summary for C^{2}U is +4.25 pp H5 Drop, -0.68 CN, -5.65 TE, indicating modest suppression with minimal utility loss. [2]
- Four-concept composition on ArXiv yields positive lift over an averaged-direction baseline after independent validation gates and first-use test evaluation. [1][3]

Limitations
- Suppression is partial: after editing, target Hit@5 remains near 90 % pooled, so concepts are reduced rather than removed. [3]
- Evaluation scope is narrow: only five English datasets, four encoders, and limited query budgets (100 target/300 control queries per unit); multi-concept claims cover only ArXiv with K = 4 and may not generalize. [3]
- The study assesses only retrieval behavior; safety of generated outputs, multilingual settings, adaptive attackers, and measured runtime costs are untested. [3]

Sources

[1] arXiv 2610.29223v1.pdf, arXiv abstract

[2] arXiv 2610.29223v1.pdf, page 6

[3] arXiv 2610.29223v1.pdf, page 10

[P4] Joint Constraint Aggregation for Qualifier-Sensitive Retrieval-Augmented QA

Jennifer Hsia, Zachary C. Lipton, Aarti Singh · arXiv 2610.27114v1 · https://arxiv.org/abs/2610.27114v1 · relevance 0.42

TL;DR: Joint Constraint Aggregation improves qualifier-sensitive answer selection in retrieval-augmented QA, especially when retrieval favors a competing near-miss answer.

Problem
Retrieval-augmented QA can select a wrong but plausible answer because retrieved passages may mention a competing answer more frequently, and candidate-level support alone cannot resolve this ambiguity [1][2].

Method
The authors introduce Joint Constraint Aggregation (JCA), which jointly compares candidate answers against the same question-derived requirements using the retrieved evidence, rather than evaluating each candidate separately; they evaluate JCA with three instruction-tuned backbones (Llama4-Scout-17B, Qwen3-30B, Gemma3-27B) across three settings (AmbigQA, AmbigQA-Extracted, SituatedQA-temporal) [1][3][4].

Key results
- With Llama4-Scout-17B, JCA attains the highest accuracy on all three full evaluation sets and on every hard-slice condition, and similar superiority is observed for Qwen3-30B and Gemma3-27B [4].
- Against the strongest baseline (CoT) the accuracy gain ranges from 4.0 to 11.3 points on full sets and from 6.1 to 15.5 points on hard slices [4].
- JCA's advantage grows as the mention gap becomes more negative, while baseline accuracies decline, showing JCA's stability across retrieval salience variations [4].
- In the AmbigQA-Extracted setting, the extraction pipeline recovers the gold answer for 68.6 % of questions, indicating an upstream bottleneck [5][7].
- Staged JCA incurs higher inference latency than single-call baselines (e.g., 4.24 s vs 1.00 s latency for CoT-thinking on AmbigQA) [8].

Limitations
- Experiments condition on candidate recall and do not assess end-to-end retrieval or generation performance, so the impact of JCA on full RAG pipelines is unknown [5].
- The hard-slice diagnostic uses alias-based mention frequency rather than semantic support, limiting its representativeness of real-world failures [5].
Accuracy gains are obtained with instruction-tuned models, and on the three hard slices staged JCA is 1-3 points higher than CoT-thinking, but these differences are not statistically significant [5].
- Staged JCA incurs higher computational cost due to multiple calls per candidate [5][8].
- The released evidence reflects a snapshot of live web search and may change over time [5].

Sources

[1] arXiv 2610.27114v1.pdf, arXiv abstract

[2] arXiv 2610.27114v1.pdf, page 3

[3] arXiv 2610.27114v1.pdf, page 4

[4] arXiv 2610.27114v1.pdf, page 6

[5] arXiv 2610.27114v1.pdf, page 9

[7] arXiv 2610.27114v1.pdf, page 11

[8] arXiv 2610.27114v1.pdf, page 17

[P5] MAPS: Macro-Adaptive Parallel Reasoning for Regulating Search Momentum in Retrieval-Augmented Generation

Xinze Li, Dingwen Zhang, Zhenghao Liu, Yukun Yan et al. · *arXiv 2610.29792v1* · *https://arxiv.org/abs/2610.29792v1* · *relevance 0.39*

TL;DR: MAPS introduces a macro-adaptive parallel reasoning framework that curbs query-intent drift in retrieval-augmented generation and yields consistent performance gains on multiple QA benchmarks.

Problem - Retrieval-augmented generation suffers from redundant or ineffective sub-queries and limited query diversity, leading to drift between generated sub-queries and the original information need [1][2]. The evolving retrieval trajectory, termed "search momentum," can cause accumulated deviations that degrade downstream reasoning performance [2].

Method - MAPS first disentangles historical reasoning trajectories by retaining retrieved knowledge and a refined reasoning state, reducing the influence of past retrieval tendencies [1][3]. It then extends sequential reasoning to a parallel paradigm: at each iteration, a search-direction proposal model (MIntent) generates multiple complementary retrieval directions, each instantiated as an independent sub-query; the resulting queries retrieve top-K documents in parallel, which are aggregated and refined into a new knowledge observation and updated reasoning state [3].

Key results - Experiments on six question-answering benchmarks show that MAPS consistently outperforms representative RAG baselines across different backbone models [4]. Specifically, MAPS improves accuracy by 6.8 % on Qwen3-32B and 5.8 % on Qwen3.6-35B-A3B, demonstrating robustness and generalizability [4].

Limitations - The provided excerpts do not contain any discussion of the paper's limitations.

Sources

[1] arXiv 2610.29792v1.pdf, arXiv abstract

[2] arXiv 2610.29792v1.pdf, page 2

[3] arXiv 2610.29792v1.pdf, page 5

[4] arXiv 2610.29792v1.pdf, page 6